

Noise Analysis — 2um

PAGE 1 — FILE STATISTICS

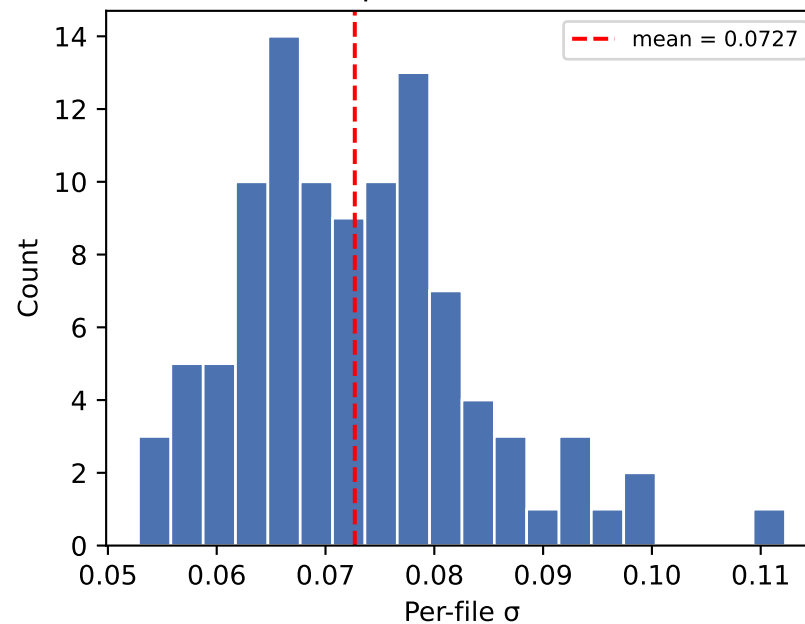
Folder: data/dataset_hamming_clean/test/2um
Files: 101
Signal length: 2500–2500 samples
Dtype: float64

Aggregate noise statistics:

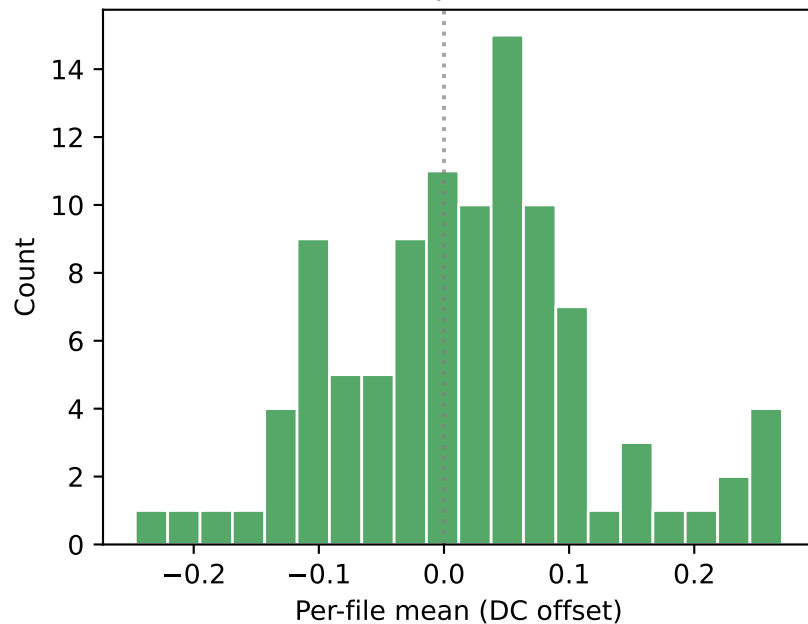
mean(σ): 0.072691
std(σ): 0.010825
CV(σ): 0.1489
mean(RMS): 0.116183
mean(kurtosis): 0.0500 ($\emptyset$ = Gaussian)

mean(DC): 0.017175
std(DC): 0.102208
Amplitude range: [-0.5288, 0.6063]

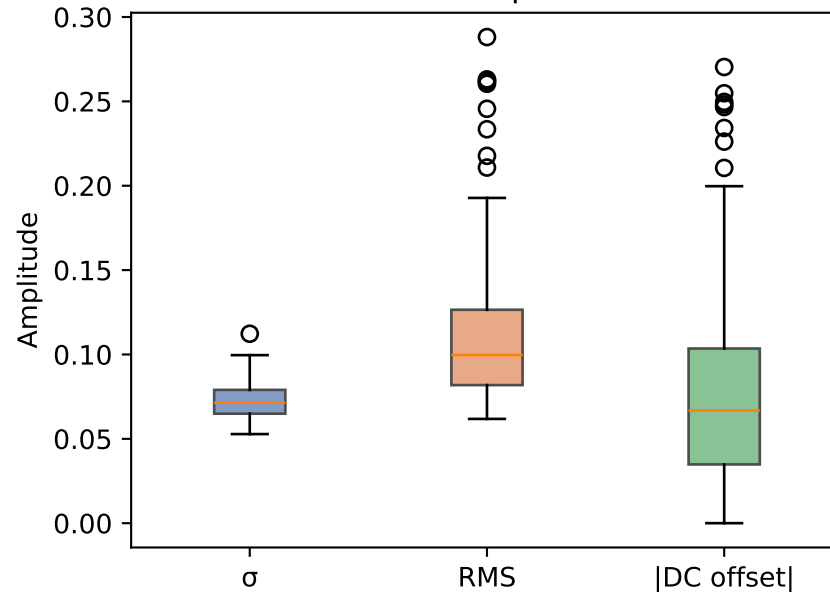
Distribution of per-file standard deviations



Distribution of per-file DC offsets

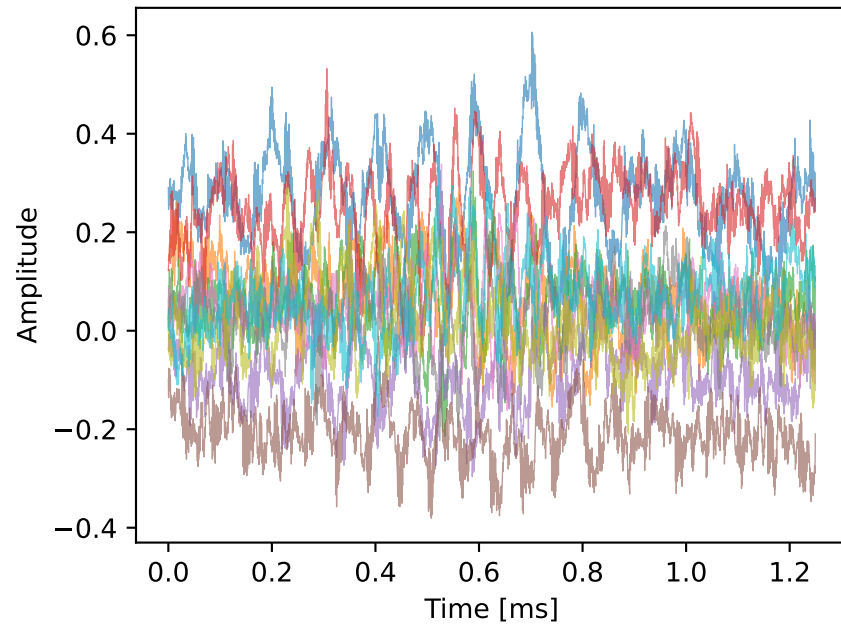


Distribution of amplitude metrics

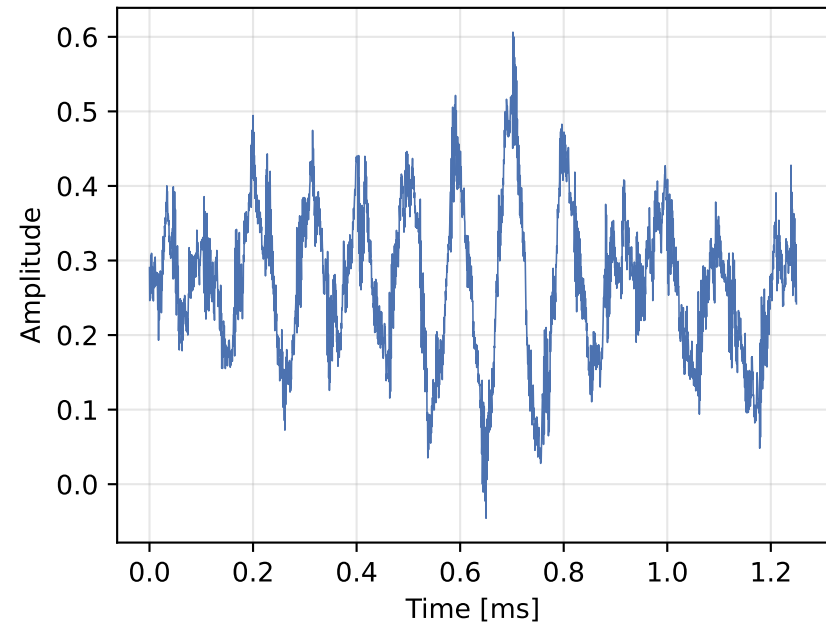


Time-Domain Analysis — 2um

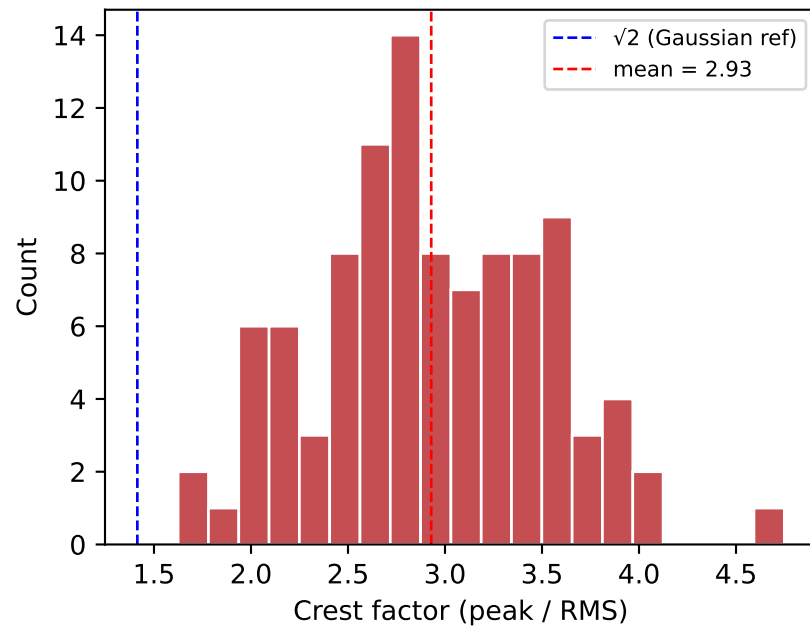
Overlay of 10 random signals



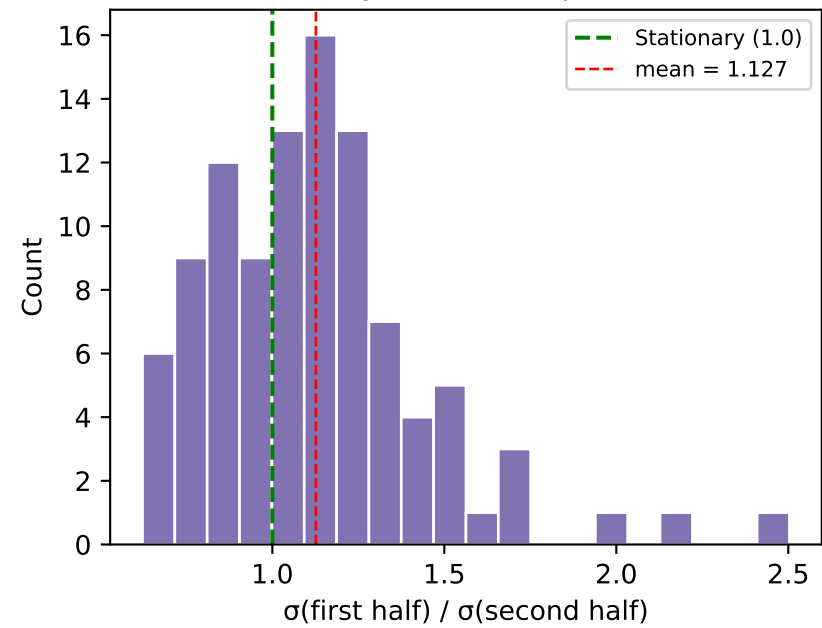
Signal #97 (detail)



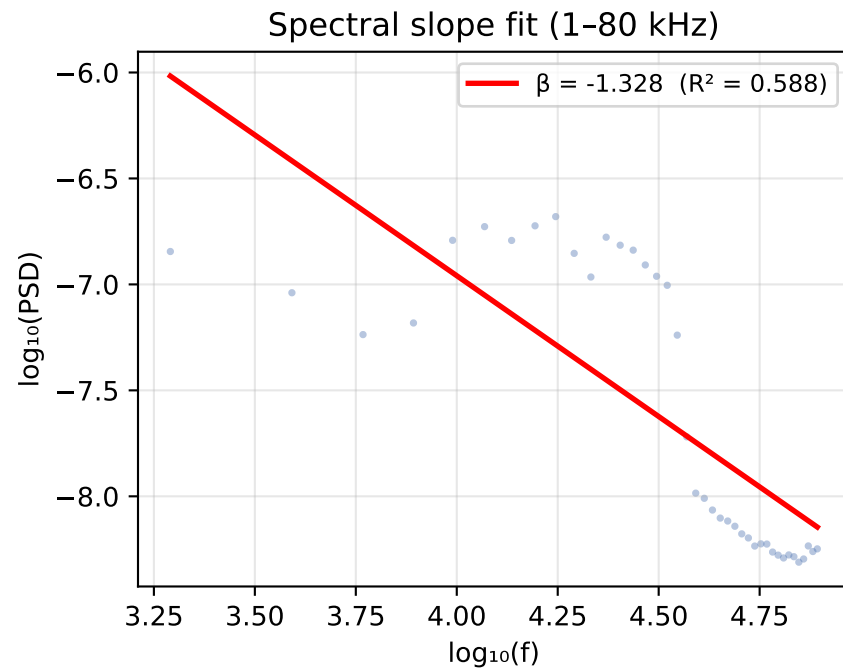
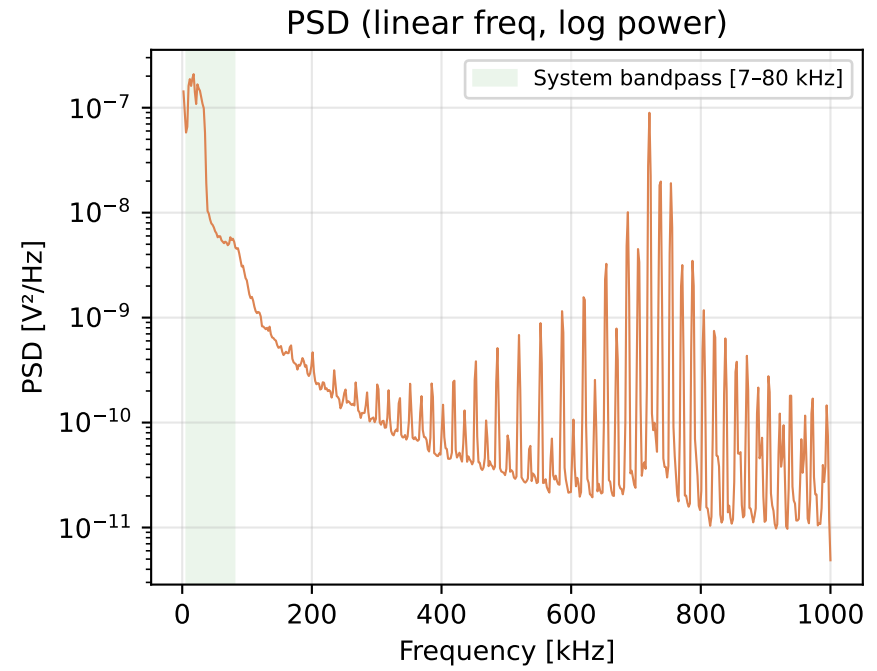
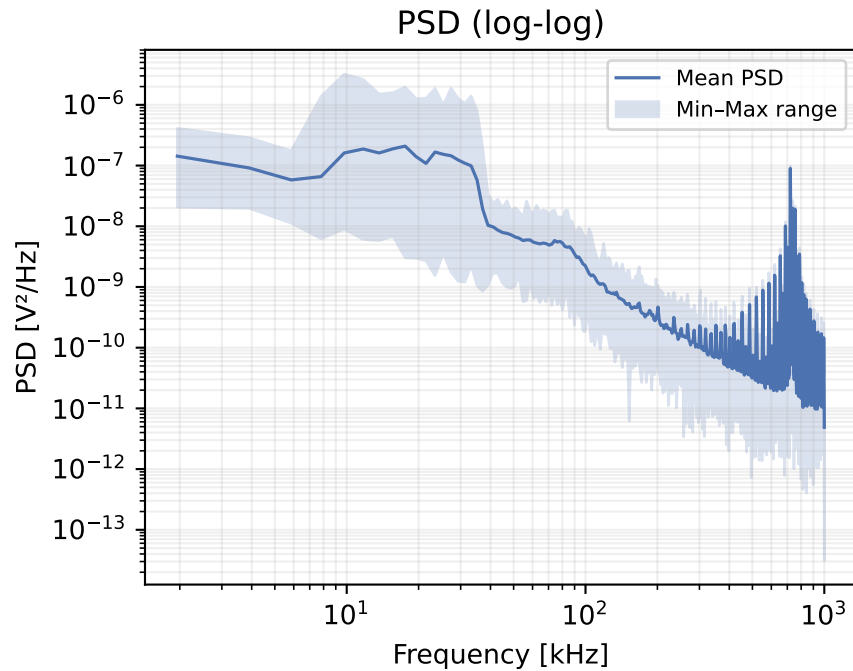
Crest factor distribution



Stationarity test (half-split σ ratio)



Power Spectral Density — 2um



SPECTRAL ANALYSIS SUMMARY

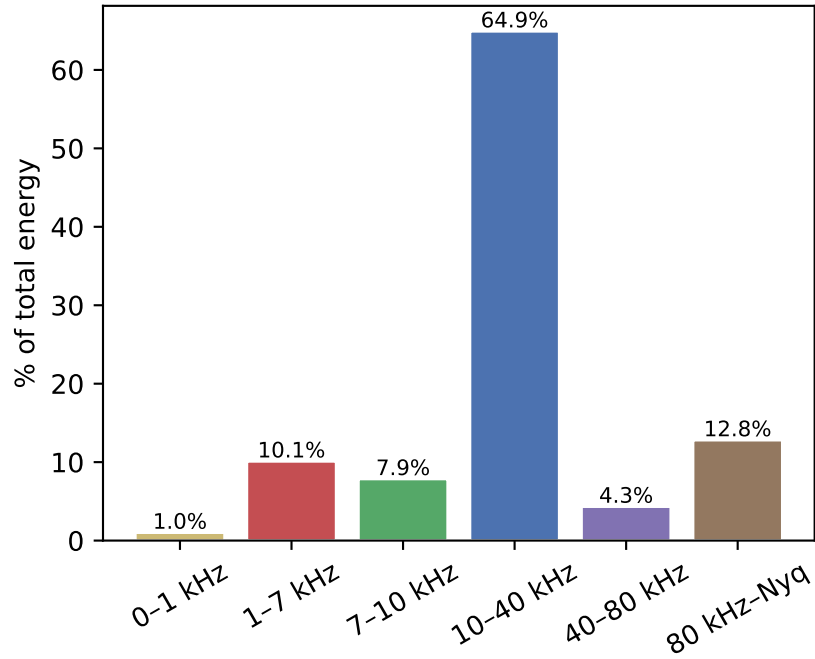
Spectral slope β : -1.3285
Fit quality R^2 : 0.5875
Spectral flatness: 0.0279 (1.0 = white)

Slope interpretation:
 $\beta \approx 0 \rightarrow$ White noise (flat PSD)
 $\beta \approx -1 \rightarrow$ Pink / $1/f$ (equal energy per octave)
 $\beta \approx -2 \rightarrow$ Brown / $1/f^2$ (random walk)

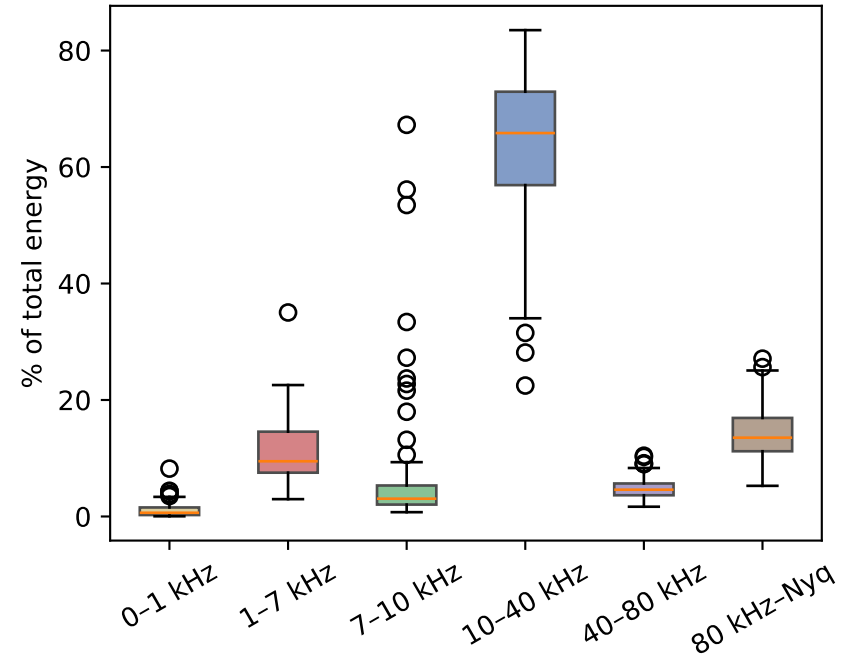
Classification: Pink ($1/f$)

Frequency Band Energy — 2um

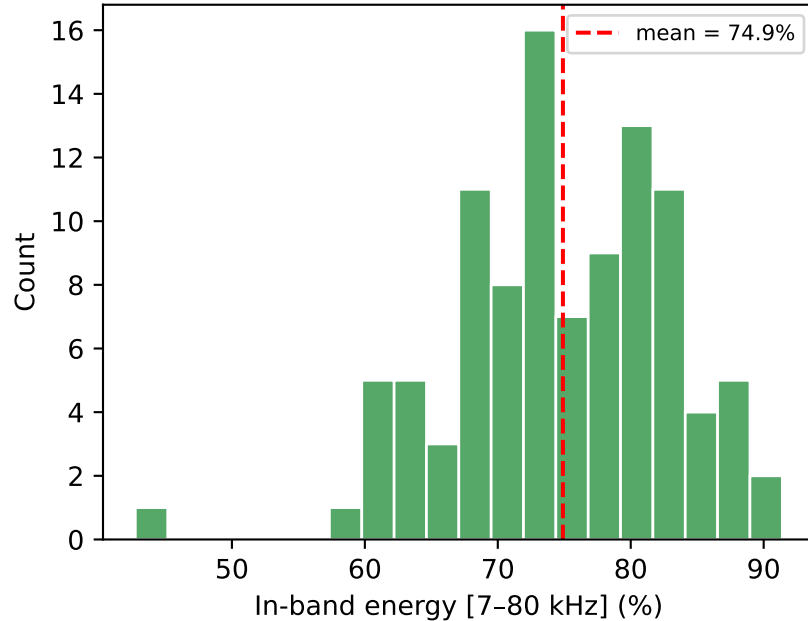
Average energy per frequency band



Per-file band energy distribution



System bandpass energy ratio per file



BAND ENERGY SUMMARY

Band	Mean %	Std %
0-1 kHz	1.0%	1.3%
1-7 kHz	10.1%	5.2%
7-10 kHz	7.9%	10.8%
10-40 kHz	64.9%	12.3%
40-80 kHz	4.3%	1.8%
80 kHz-Nyq	12.8%	4.6%
In-band [7-80k]	74.9%	8.2%

Noise Type Classification — 2um

NOISE TYPE CLASSIFICATION

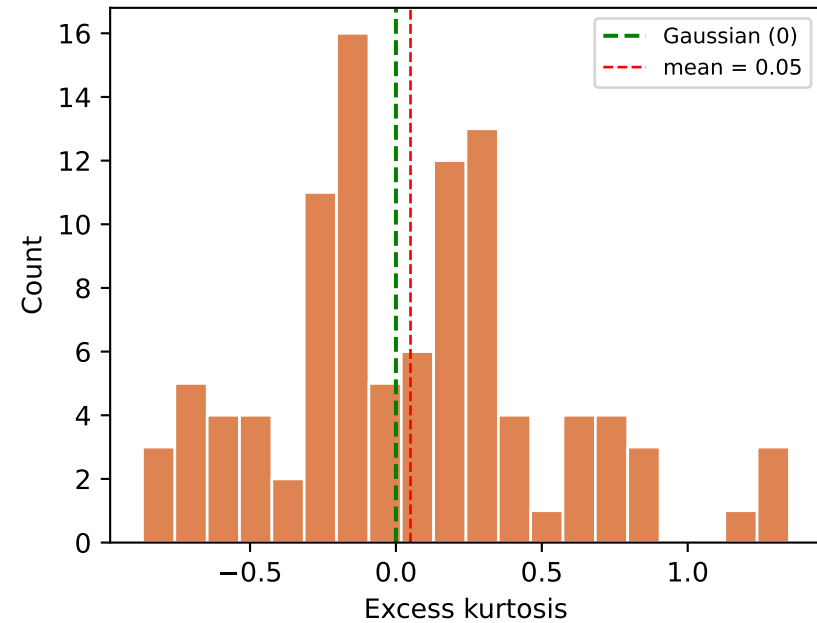
Spectral color: Pink (1/f)
slope β = -1.3285
flatness = 0.0279

Amplitude dist.: Gaussian
kurtosis (mean): 0.0500
(excess kurtosis: 0 = Gaussian,
>0 = heavy-tailed, <0 = light-tailed)

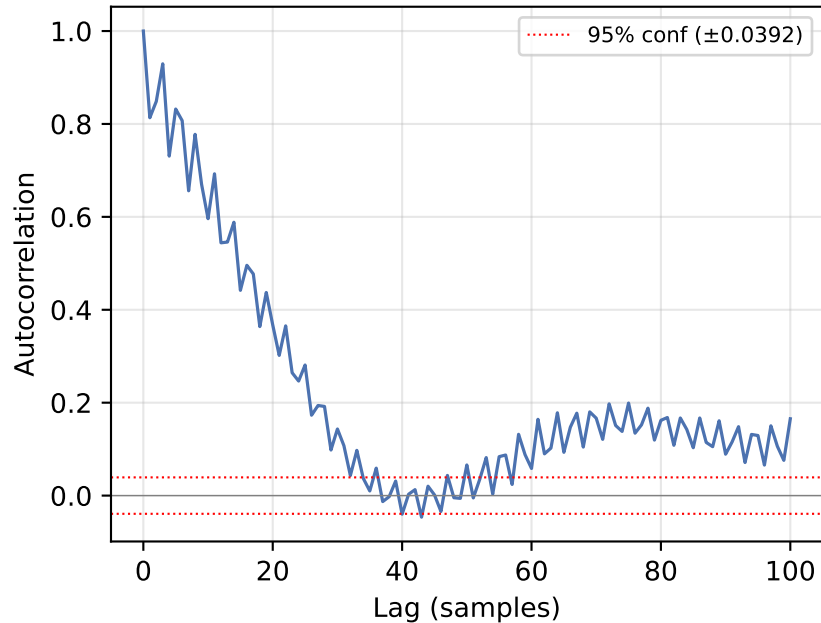
Decision boundaries:
 $|\beta| < 0.3$ → White
0.3–1.5 → Pink (1/f)
> 1.5 → Brown (1/f²)

 $|\text{kurt}| < 0.5$ → Gaussian
0.5–2.0 → Near-Gaussian
> 2.0 → Non-Gaussian

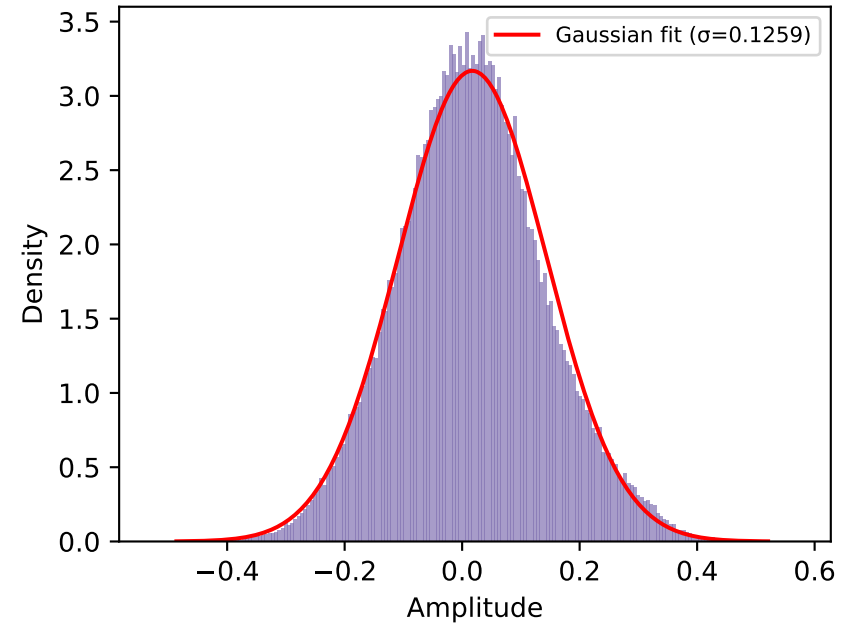
Per-file kurtosis distribution



Average autocorrelation (first 100 lags)

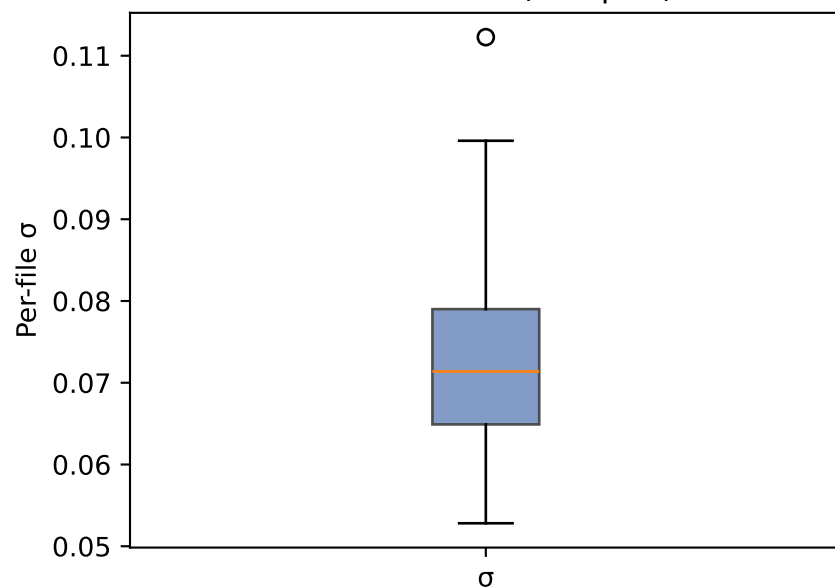


Amplitude distribution (all samples pooled)

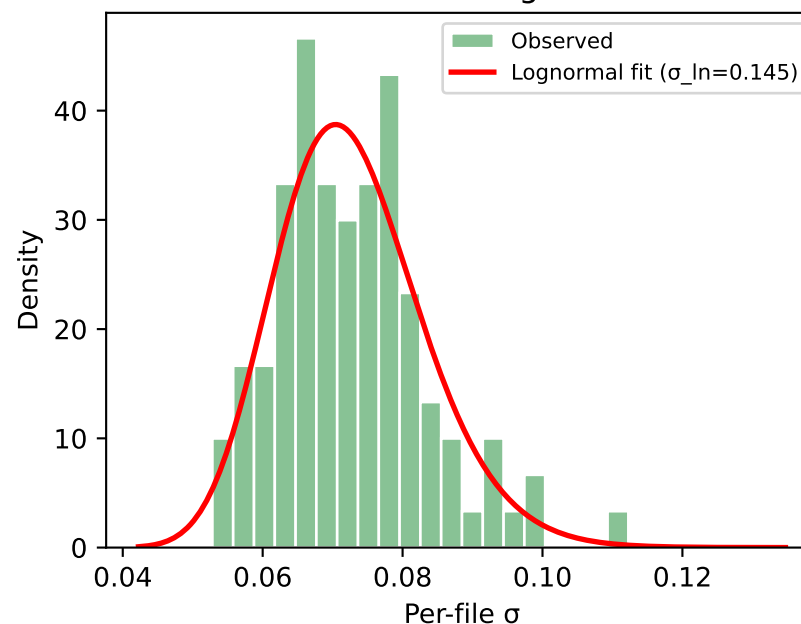


Amplitude Variability (Inter-file) — 2um

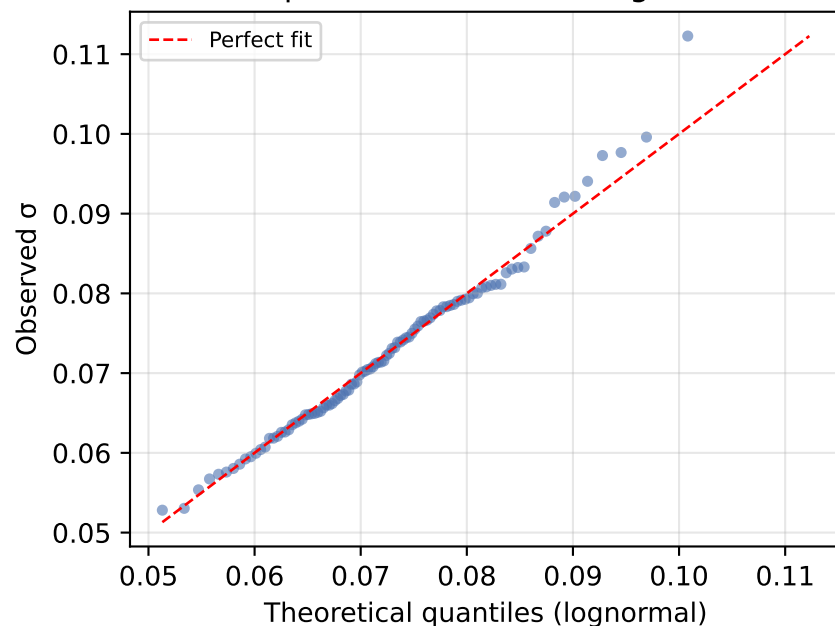
σ distribution (box plot)



σ distribution + lognormal fit



Q-Q plot: observed σ vs lognormal



AMPLITUDE VARIABILITY SUMMARY

Number of files: 101
mean(σ): 0.072691
std(σ): 0.010825
CV = std(σ)/mean(σ): 0.1489

Lognormal fit parameters:
shape ($\sigma_{\ln}$): 0.1447
scale ($\exp(\mu_{\ln})$): 0.071922

Reference (from real noise files):
CV_ref = 0.19
Ratio CV/CV_ref: 0.78

Interpretation:
Moderate variability